

SOFTWARE ENGINEERING

ASSIGNMENT 1

SESSION 1

1. Name 5 IT job roles and write one line about what each role does.

➔ **1. Front-End Developer**

Builds the user interface of websites and applications that users see and interact with.

2. Back-End Developer

Develops the server-side logic, databases, and APIs that power applications behind the scenes.

3. Full-Stack Developer

Works on both the front-end and back-end parts of an application.

4. QA Tester

Tests software to find bugs and ensure quality before release.

5. Data Analyst

Analyzes data and creates reports to support business decisions.

2. List 3 product companies and 3 service companies with one example of work each does.

➔ **Product Companies**

1. **Microsoft** – Develops the Windows operating system and Office products.

2. **Google** – Creates products such as Android and Google Search.

3. **Adobe** – Develops software like Photoshop and Premiere Pro.

Service Companies

1. **Tata Consultancy Services** – Provides software development and IT consulting services.

2. **Infosys** – Offers application development and digital transformation services.

3. **Wipro** – Provides cloud, cybersecurity, and IT support services.

3. Write the full forms and one-line use of: **GitHub, Slack, Jira, IDE, API.**

➔ **1. GitHub** - [**Git** (a distributed version control tool) + **Hub** (a central hosting platform)
= **GitHub** (Git-based code hosting platform)]

Used to store, manage, and collaborate on source code.

2. Slack - Searchable Log of All Communication and Knowledge

Used for team communication and collaboration.

3. Jira - Not officially an acronym

Used for project management and bug tracking.

4. IDE - Integrated Development Environment

Used to write, run, and debug code.

5. API - Application Programming Interface

Allows different software applications to communicate with each other.

4. Name any 5 programming languages and mention one popular use case for each.

➔ **1. Python** – Machine Learning

2. JavaScript – Web Development

3. Java – Android App Development

4. C++ – Game development

5. PHP – Backend Web Development

5. What is the difference between a developer and a tester? Answer in 2 lines.

➔ A **developer** writes and maintains the code that makes a software application work.

A **tester** checks application for bugs, verifies, ensures the software works correctly before release.